

SIMATS ENGINEERING (SAVEETHA SCHOOL OF ENGINEERING)

Department of Computer Science and Engineering

ASSESSMENT TOOL 2 – SCENARIO BASED ASSESSMENT

Course Code:	CSA0615	Course Title:	Design and Analysis of Algorithms
CO Assessed:	CO2 – Manipulation (BL1, BL2, BL3)	Assessment:	Assessment Tool 2 – Scenario Based Assessment
Weightage:	20% of CIA	Total Marks:	100
CO2 Statement:	Apply Brute Force technique to solve sorting, searching, Closest-Pair and Convex-Hull problems (BL1, BL2, BL3)		
Student Name:	SOUNDARIYA M	Reg. No.:	192524342
Section / Batch:	2nd Year	Date:	31-07-2026

Instructions to Students

- Answer BOTH scenario-based questions. Each question carries 50 Marks. Total: 100 Marks.
- Carefully analyze the given scenario and identify the computational problem involved.
- Apply appropriate Brute Force techniques for sorting, searching, Closest-Pair, and Convex-Hull problems wherever applicable.
- Clearly justify the chosen solution approach with proper reasoning and analytical interpretation.
- Demonstrate decision-making skills by comparing possible solutions and selecting the most suitable approach.
- Use diagrams, stepwise illustrations, or algorithmic representations wherever necessary.
- Maintain clarity, logical organization, and proper technical presentation throughout the answers.

Question Type	Focus Area	Questions	Marks
Scenario Based Assessment	Analyze real-world scenarios and apply Brute Force techniques for sorting, searching, Closest-Pair, and Convex-Hull problems	Q1 - Q2	100 (2 × 50)
GRAND TOTAL:		100 Marks	

SECTION A – SCENARIO BASED ASSESSMENT

1.



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Soundariya M
192524342RA

Q1
Brute Force
100 marks

Q2
Brute Force
100 marks

1/2 answered

A text editor searches for a keyword within large documents using brute-force string matching. As document size increases, search time becomes significantly high. a) Explain how brute-force string matching works in this context.
b) Analyze the time complexity of the algorithm.
c) Suggest why this approach is inefficient and justify the need for optimized algorithms.

Your Code

```
1 text=input("enter the text:")
2 pattern=input("enter the pattern:")
3 n=len(text)
4 m=len(pattern)
5 found=False
6 for i in range(n-m+1):
7     j=0
8     while j<m and text[i+j]==pattern[j]:
9         j+=1
10    if j==m:
11        print("pattern found at position:",i+1)
12        found=True
13 if not found:
14    print("pattern not found")
```

Run Save Submit

Input

artificial intelligence
intelligence

Output

enter the text:enter the
pattern:pattern found at position:
12

2.

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192524342RA

Q1
Brute Force
100 marks

Q2
Brute Force
100 marks

2/2 answered

A financial system maintains sorted transaction data for quick lookup using Binary Search, but frequent updates disrupt the order. a) Explain how Binary Search operates on sorted data.

b) Analyze its performance when order is maintained versus disturbed.

c) Discuss the importance of maintaining sorted data.

Your Code

```

1 arr=list(map(int,input("enter sorted numbers separated by space:").split()))
2 key=int(input("enter element to search:"))
3 low=0
4 high=len(arr)-1
5 found=False
6 while low<=high:
7     mid=(low+high)//2
8     if arr[mid]==key:
9         print("element found at position:",mid+1)
10        found=True
11        break
12    elif arr[mid]<key:
13        low=mid+1
14    else:
15        high=mid-1
16 if not found:
17     print("element not found")
                
```

Input

10 20 30 40 50 60 70
50

Output

enter sorted numbers separated by
space:/nenter element to
search:element found at position: 5

Code of Conduct

I SOUNDAIRIYA M(192524342) (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment. I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the Student: **Soundariya M**

RUBRICS FOR CALCULATION

Criteria	Wt.	Level 4 – Exemplary	Level 3 – Proficient	Level 2 – Developing	Level 1 – Beginning
Problem Understanding & Context Analysis	25	Thoroughly understands the scenario with accurate interpretation of all requirements	Clearly understands the scenario with minor gaps	Basic understanding; some aspects of the scenario unclear	Poor understanding or misinterpretation of the scenario
Application of Concepts	30	Applies appropriate concepts effectively to solve complex real-world problems	Applies relevant concepts correctly with minor errors	Applies basic concepts but with limited accuracy	Fails to apply appropriate concepts
Analytical & Decision-Making Skills	20	Demonstrates strong analysis and selects optimal solutions with justification	Good analysis with reasonable solutions	Limited analysis; solutions lack justification	Poor or no analysis; incorrect decisions

Solution Design & Approach	15	Well-structured, innovative, and feasible solution approach	Logical and structured solution with minor gaps	Basic solution with limited structure or feasibility	Unclear or incorrect solution approach
Clarity, Justification & Presentation	10	Clear, well-justified explanation with strong reasoning	Clear explanation with some justification	Limited clarity and weak justification	Poorly explained with no justification

Marks Evaluation:

Question No.	Problem Understanding & Context Analysis (25%)	Application of Concepts (30%)	Analytical & Decision-Making Skills (20%)	Solution Design & Approach (15%)	Clarity, Justification & Presentation (10%)	Total Marks (Each – 50)
Q1	4	3	3	2	4	40
Q2	4	4	4	3	2	46
Overall Total (100)						86

Faculty In-charge	Course Coordinator	Head of Department
Signature: _____	Signature: _____	Signature: _____
Date: _____	Date: _____	Date: _____